

Problem 1: Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for each of the following:

- a. $\mathbf{F} = [2y, 3z, x]$ and C : consists of the straight paths from $(0, 2, 0)$ to $(2, 2, 0)$ to $(2, 2, 1)$ and back to $(0, 2, 0)$
- b. $\mathbf{F} = [2y, 3z, x]$ and C : consists of the straight paths from $(0, 0, 0)$ to $(0, 2, 0)$ to $(1, 1, 1)$ and back to $(0, 0, 0)$
- c. $\mathbf{F} = y\mathbf{i} + (2 - 3yz)\mathbf{j} + (x^2 + y^2)\mathbf{k}$ and C is the boundary of the side of the cylinder $x^2 + z^2 = 16$ in the first octant with $0 \leq y \leq 2$.